

Prabal Batra

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EDUCATION

Chandigarh University

B.E. in Computer Science & Engineering in Big Data and Analytics

CGPA : 7.09 (3rd Year)

Gharuan , Mohali

08/2022 – Present

WORK EXPERIENCE

• Genesys International Corporation Limited:

Apprenticeship Trainee

Mumbai, India

June 2025 - Present

• Ozibook:

Business Analyst Assistant Intern

Bangalore, India

June 2024 - Aug 2024

PROJECT EXPERIENCE

Agentic RAG-Based Geospatial Intelligence Platform :-

02/2026 – Present

- Designed and deployed a production-grade GenAI system using Agentic / Structured RAG, enabling accurate geospatial query understanding and processing 100+ spatial queries/day via PostGIS/APIs.
- Built **multi-agent LLM workflows** for intent parsing, task decomposition, and execution, improving query handling accuracy by **~40% for complex spatial queries**.
- Engineered a **custom retrieval pipeline (non-vector RAG)** with context filtering, reducing LLM input size from **~12,000 to ~900 tokens (~92% reduction)** and lowering response latency by **~60%**.

Multi-Agent Research & Reasoning System with Graph-Based RAG :-

06/2025 – 01/2026

- Developed a multi-agent GenAI system using LangChain and LangGraph, enabling task planning and execution workflows that improved multi-step query handling efficiency by **~40%**
- Implemented Retrieval-Augmented Generation (RAG) using FAISS/Chroma vector embeddings, achieving **~30–50%** improvement in response relevance for unstructured data queries
- Designed graph-based reasoning pipelines with specialized agents (research, analysis, summarization), reducing response generation time by **~35%** through parallelized task execution

Astroterra – Lunar Crater Detection & Autonomous Navigation System :-

12/2024 – 05/2025

- Developed a computer vision pipeline using CNN-based transfer learning to detect lunar craters from high-resolution satellite imagery, achieving **~85–90% detection accuracy**
- Implemented image tiling (256×256 patches) to process large-scale imagery, enabling handling of **1000+ images** and improving training efficiency by **~50%**
- Designed bounding box overlap resolution using grid-based spatial indexing, reducing duplicate detections by **~30–40%** in dense crater regions

TECHNICAL SKILLS

- Frameworks & Platforms:** LangChain, LangGraph, TensorFlow, PyTorch, Scikit-learn, FastAPI, Jupyter Notebook, Google Colab, AWS, Docker, Git, PyCharm.
- AI/ML Techniques:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG – Structured & Vector-based), Prompt Engineering, Multi-Agent Systems, Task Decomposition & Planning, Natural Language Processing (NLP), Machine Learning, Deep Learning, Computer Vision.
- Data Analysis & Processing:** Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Statistical Analysis, Context Filtering & Data Preprocessing for LLM Pipelines.
- Software Development:** Python, Data Structures and Algorithms (DSA), Object-Oriented Programming (OOPs), API Development (FastAPI), LLM Orchestration & Backend System Design.
- Tools:** NumPy, Pandas, OpenCV, MySQL, Power BI, Tableau.
- Domain:** Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Generative AI.

ACHIEVEMENTS

- Awarded **Academic Excellence Award**, Chandigarh University, for outstanding academic performance.
- Filed and published **2 patent applications** demonstrating innovation across domains:
 - Artificial Intelligence Enabled Tracking-Based Security System (Application No: 202311054397).
 - Mycofibre Composite and Method for Synthesis Thereof (Application No: 202311048967).
- Published a **book chapter** in “Hybrid Metal Additive Manufacturing Technology and Application (2023).
- At **Genesys International**, presented an **Agentic AI system** to leadership and investors; contributed to multiple **project bidding presentations**.